



Surface Area and Enlargement

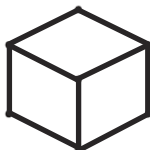
I can solve problems involving 3D shapes where the scale factor is known.



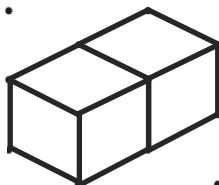
You can use interlocking cubes to help you answer these questions.

1. On the dotted paper, enlarge these shapes by a scale factor of two:

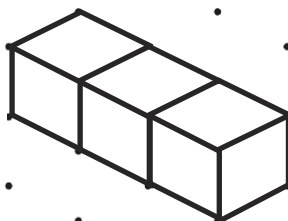
a.



b.



c.





Surface Area and Enlargement

2. What is the surface area of each shape in question one, before and after enlargement?

Shape	Before Enlargement	After enlargement
a.		
b.		
c.		



Surface Area and Enlargement

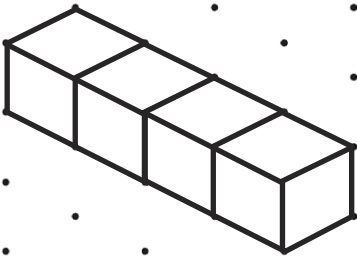
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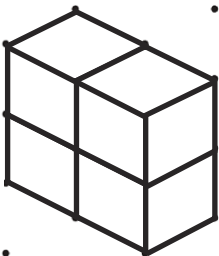
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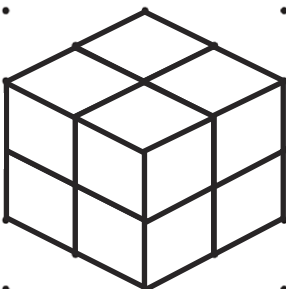
a.



b.



c.





Surface Area and Enlargement

2. What is the surface area of each shape in question one, before and after enlargement?

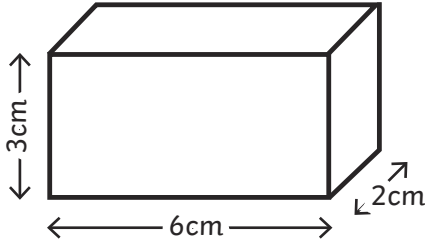
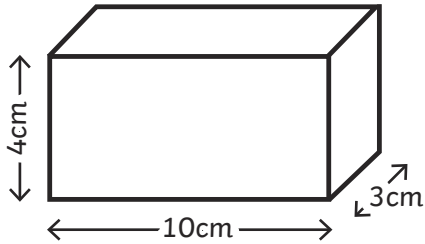
Shape	Before Enlargement	After enlargement
a.		
b.		
c.		

3. What do you notice about the relationship between the surface area of the shape before and after it has been enlarged?



Surface Area and Enlargement

4. What is the surface area of these cuboids before and after enlargement? Show your working out:

Shape	Before Enlargement	After enlargement
a.  Enlarge by a scale factor of two		
b.  Enlarge by a scale factor of two		



Surface Area and Enlargement

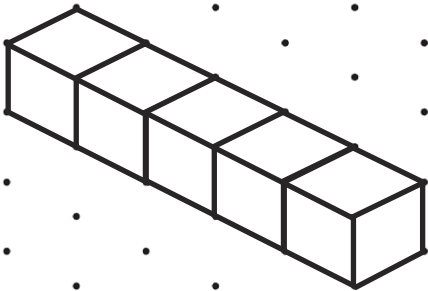
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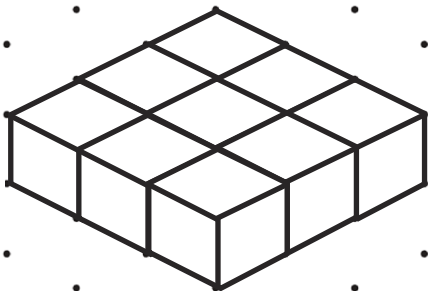
You can use interlocking cubes to help you answer these questions.

1. On the dotted paper, enlarge these shapes by a scale factor of two:

a.



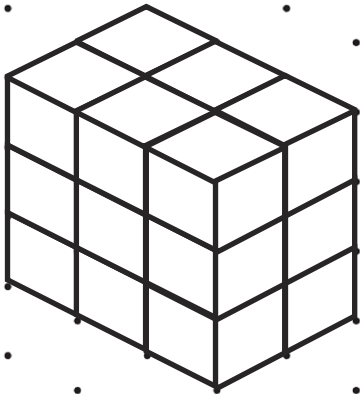
b.

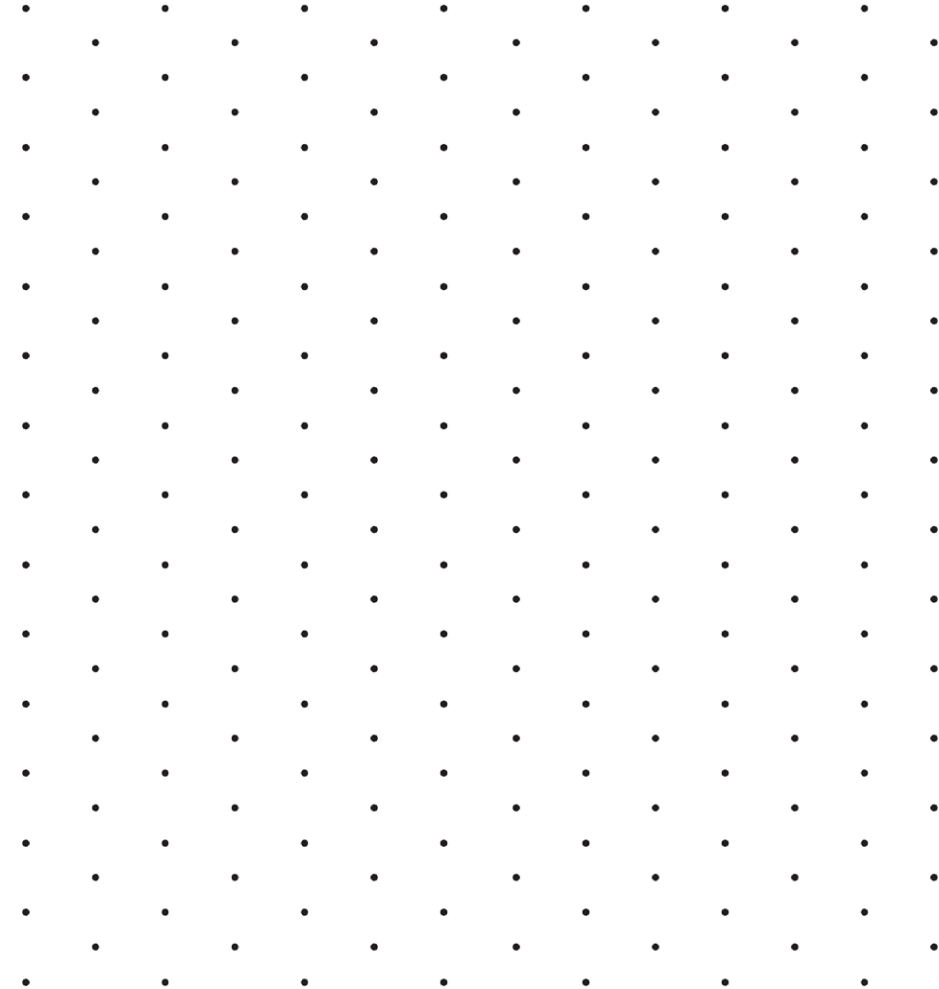




Surface Area and Enlargement

c.





2. What is the surface area of each shape in question one, before and after enlargement?

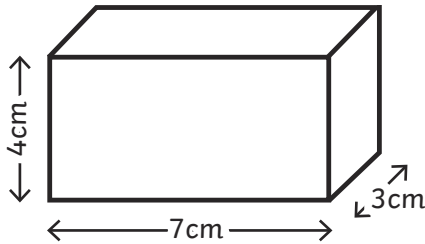
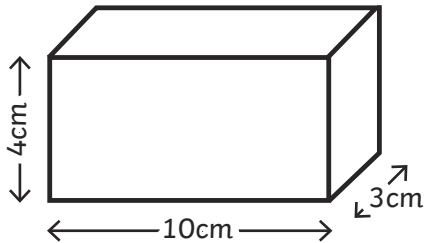
Shape	Before Enlargement	After enlargement
a.		
b.		
c.		



Surface Area and Enlargement

3. What do you notice about the relationship between the surface area of the shape before and after it has been enlarged?

4. What is the surface area of these cuboids before and after enlargement? Show your working out:

Shape	Before Enlargement	After enlargement
a.  Enlarge by a scale factor of two		
b.  Enlarge by a scale factor of three		



Surface Area and Enlargement - Answers

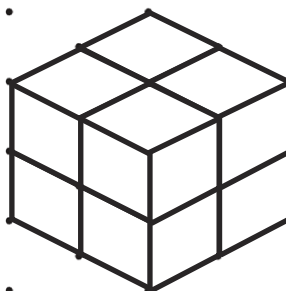
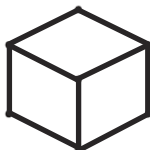
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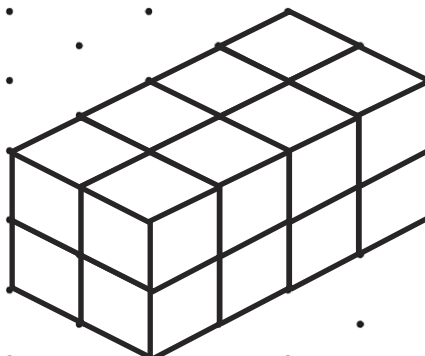
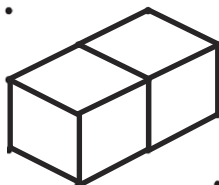
You can use interlocking cubes to help you answer these questions.

1. On the dotted paper, enlarge these shapes by a scale factor of two:

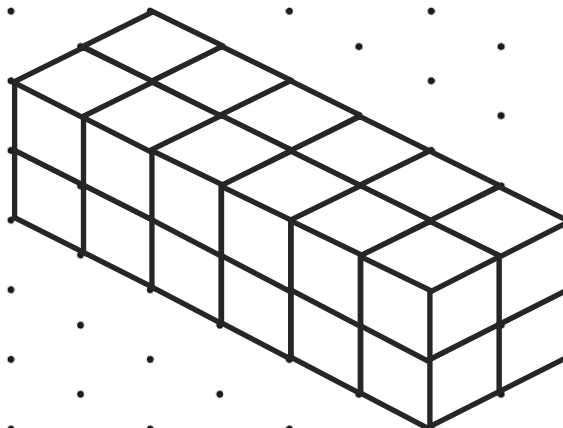
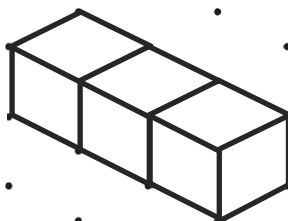
a.



b.



c.





Surface Area and Enlargement - **Answers**

2. What is the surface area of each shape in question one, before and after enlargement?

Shape	Before Enlargement	After enlargement
a.	<i>6 squares</i>	<i>24 squares</i>
b.	<i>10 squares</i>	<i>40 squares</i>
c.	<i>14 squares</i>	<i>56 squares</i>



Surface Area and Enlargement - Answers

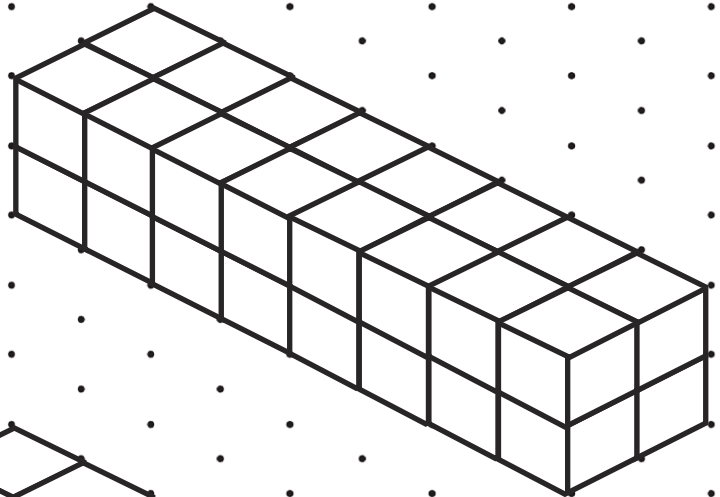
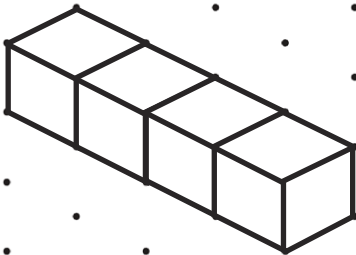
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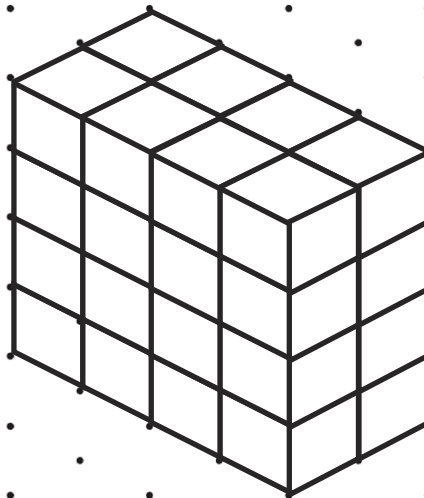
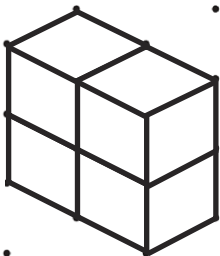
You can use interlocking cubes to help you answer these questions.

1. On the dotted paper, enlarge these shapes by a scale factor of two:

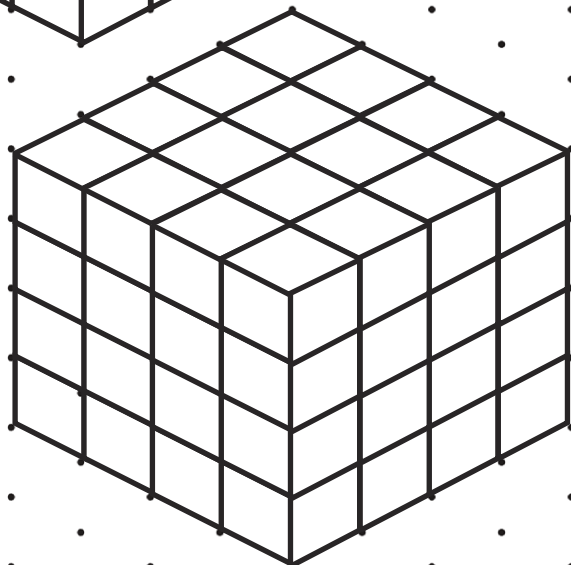
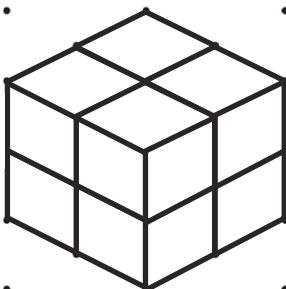
a.



b.



c.





Surface Area and Enlargement - Answers

2. What is the surface area of each shape in question one, before and after enlargement?

Shape	Before Enlargement	After enlargement
a.	<i>18 squares</i>	<i>72 squares</i>
b.	<i>16 squares</i>	<i>64 squares</i>
c.	<i>24 squares</i>	<i>96 squares</i>

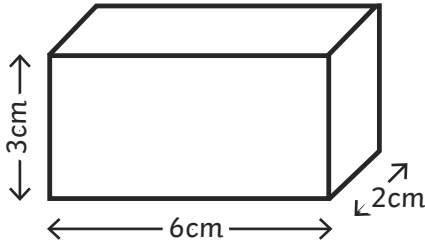
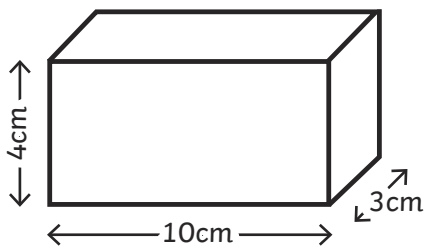
3. What do you notice about the relationship between the surface area of the shape before and after it has been enlarged?

Accept an answer which show understanding that the surface area after enlarging by a scale factor of two is multiplied by four.



Surface Area and Enlargement - Answers

4. What is the surface area of these cuboids before and after enlargement? Show your working out:

Shape	Before Enlargement	After enlargement
a.  Enlarge by a scale factor of two	84 cm² <i>Working out shows that the larger faces were calculated by multiplying 6×3 then by 4 = 72, and the 2 smaller faces by multiplying 3×2 then by 2 = 12. Then adding 72 and 12.</i> <i>Repeated addition is also acceptable.</i>	336 cm² <i>Either multiply by 4 the answer for pre enlargement – $84 \times 4 = 336$</i> <i>Or multiply the dimensions by 2:</i> <i>$6\text{cm} \times 2 = 12\text{cm}$; $3\text{cm} \times 2 = 6\text{cm}$; $2\text{cm} \times 2 = 4\text{cm}$</i> <i>Larger faces = $12 \times 6 = 72$ then multiply by 4 = 288</i> <i>Smaller faces = $6 \times 4 = 24$ then multiply by 2 = 48. Then adding 288 and 48.</i>
b.  Enlarge by a scale factor of two	184 cm² <i>Working out shows that the larger faces were calculated by multiplying 10×4 then by 4 = 160, and the 2 smaller faces by multiplying 4×3 then by 2 = 24. Then adding 160 and 24.</i> <i>Repeated addition is also acceptable.</i>	736 cm² <i>Either multiply by 4 the answer for pre enlargement – $184 \times 4 = 736$</i> <i>Or multiply the dimensions by 2:</i> <i>$10\text{cm} \times 2 = 20\text{cm}$; $4\text{cm} \times 2 = 8\text{cm}$; $3\text{cm} \times 2 = 6\text{cm}$</i> <i>Larger faces = $20 \times 8 = 160$ then multiply by 4 = 640</i> <i>Smaller faces = $8 \times 6 = 48$ then multiply by 2 = 96. Then adding 640 and 96.</i>



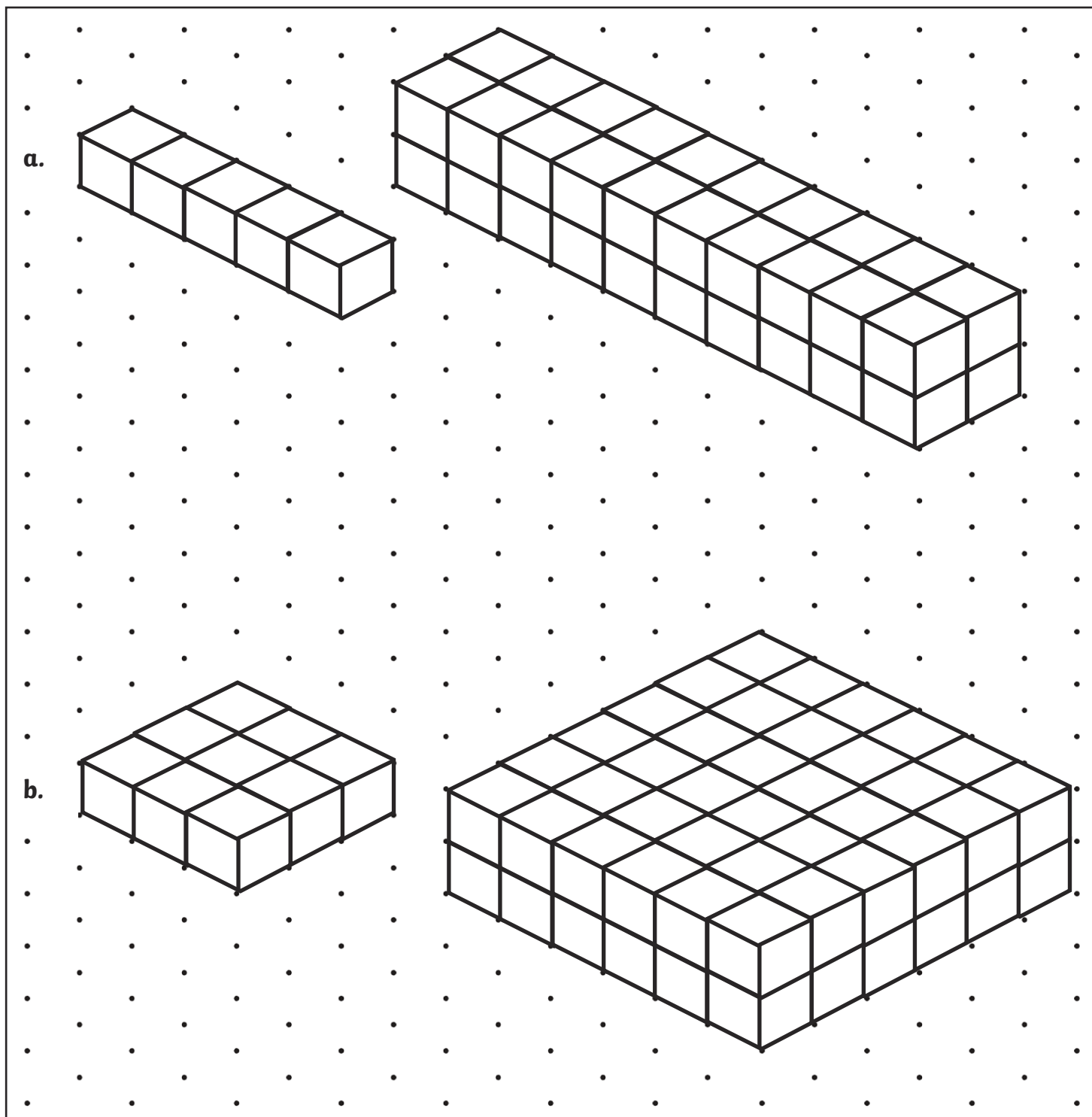
Surface Area and Enlargement - Answers

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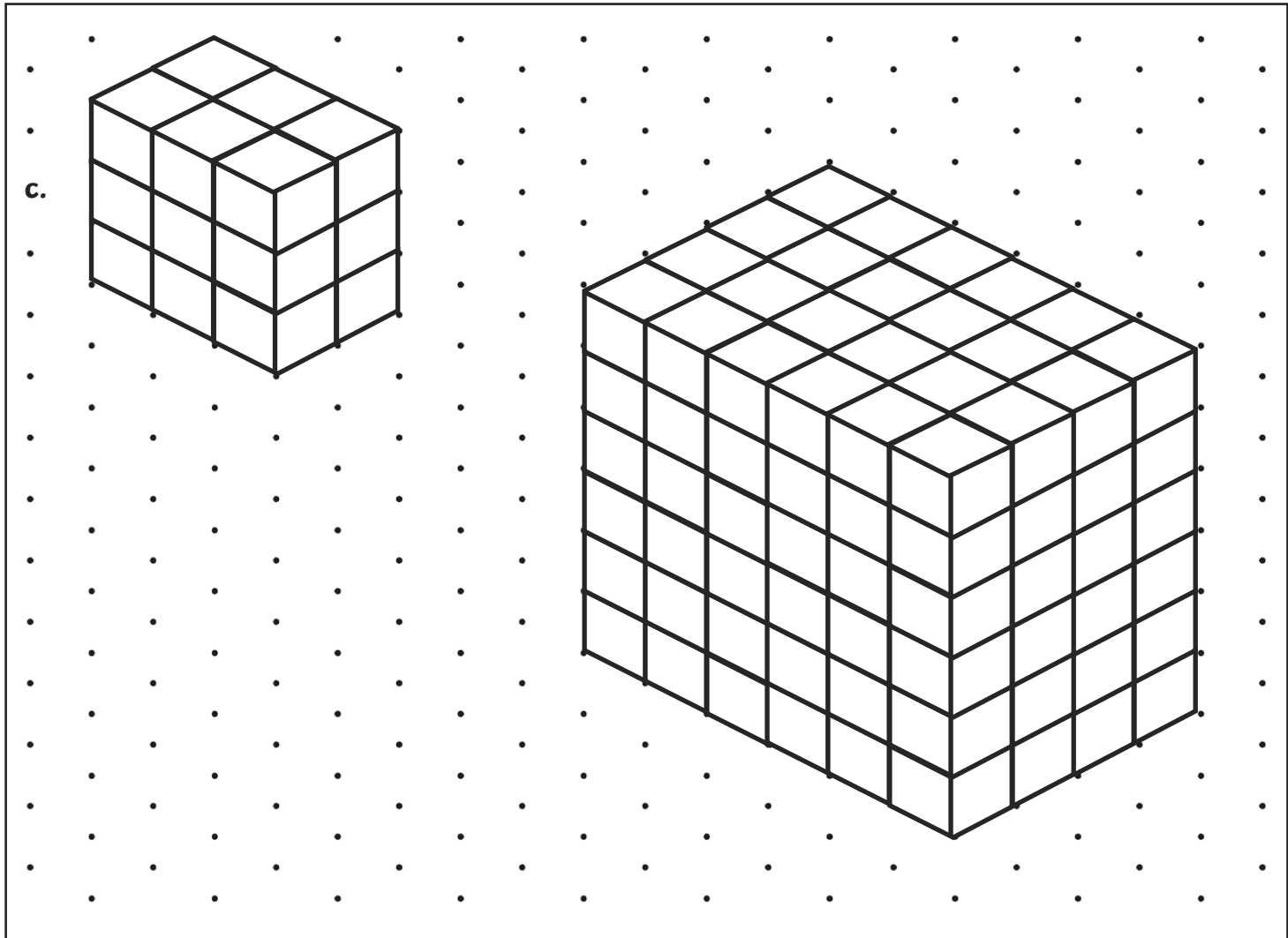
You can use interlocking cubes to help you answer these questions.

1. On the dotted paper, enlarge these shapes by a scale factor of two:





Surface Area and Enlargement - **Answers**



2. What is the surface area of each shape in question one, before and after enlargement?

Shape	Before Enlargement	After enlargement
a.	22 squares	88 squares
b.	30 squares	120 squares
c.	42 squares	168 squares



Surface Area and Enlargement - Answers

3. What do you notice about the relationship between the surface area of the shape before and after it has been enlarged?

Accept an answer which show understanding that the surface area after enlarging by a scale factor of two is multiplied by four.

4. What is the surface area of these cuboids before and after enlargement? Show your working out:

Shape	Before Enlargement	After enlargement
a. Enlarge by a scale factor of two	136 cm^2 Working out shows that the larger faces were calculated by multiplying 7×4 then by 4 = 112, and the 2 smaller faces by multiplying 4×3 then by 2 = 24. Then adding 112 and 24. Repeated addition is also acceptable.	544 cm^2 Either multiply by 4 the answer for pre enlargement – $136 \times 4 = 544$ Or multiply the dimensions by 2: $7\text{cm} \times 2 = 14\text{cm}$; $4\text{cm} \times 2 = 8\text{cm}$; $3\text{cm} \times 2 = 6\text{cm}$ Larger faces = $14 \times 8 = 112$ then multiply by 4 = 448 Smaller faces = $8 \times 6 = 48$ then multiply by 2 = 96. Then adding 448 and 96.
b. Enlarge by a scale factor of three	184 cm^2 Working out shows that the larger faces were calculated by multiplying 10×4 then by 4 = 160, and the 2 smaller faces by multiplying 4×3 then by 2 = 24. Then adding 160 and 24. Repeated addition is also acceptable.	1656 cm^2 Either multiply by 9 the answer for pre enlargement – $184 \times 9 = 1656$ Or multiply the dimensions by 3: $10\text{cm} \times 3 = 30\text{cm}$; $4\text{cm} \times 3 = 12\text{cm}$; $3\text{cm} \times 3 = 9\text{cm}$ Larger faces = $30 \times 12 = 360$ then multiply by 4 = 1440 Smaller faces = $12 \times 9 = 108$ then multiply by 2 = 216. Then adding 1440 and 216.